Supplementary Information for:

Chapter 3 | Genomic Status of the Eurasian Curlew *Numenius arquata*: Estimating Essential Biodiversity Variables and Selection Signals for a Declining Migratory Bird

This PDF file includes:

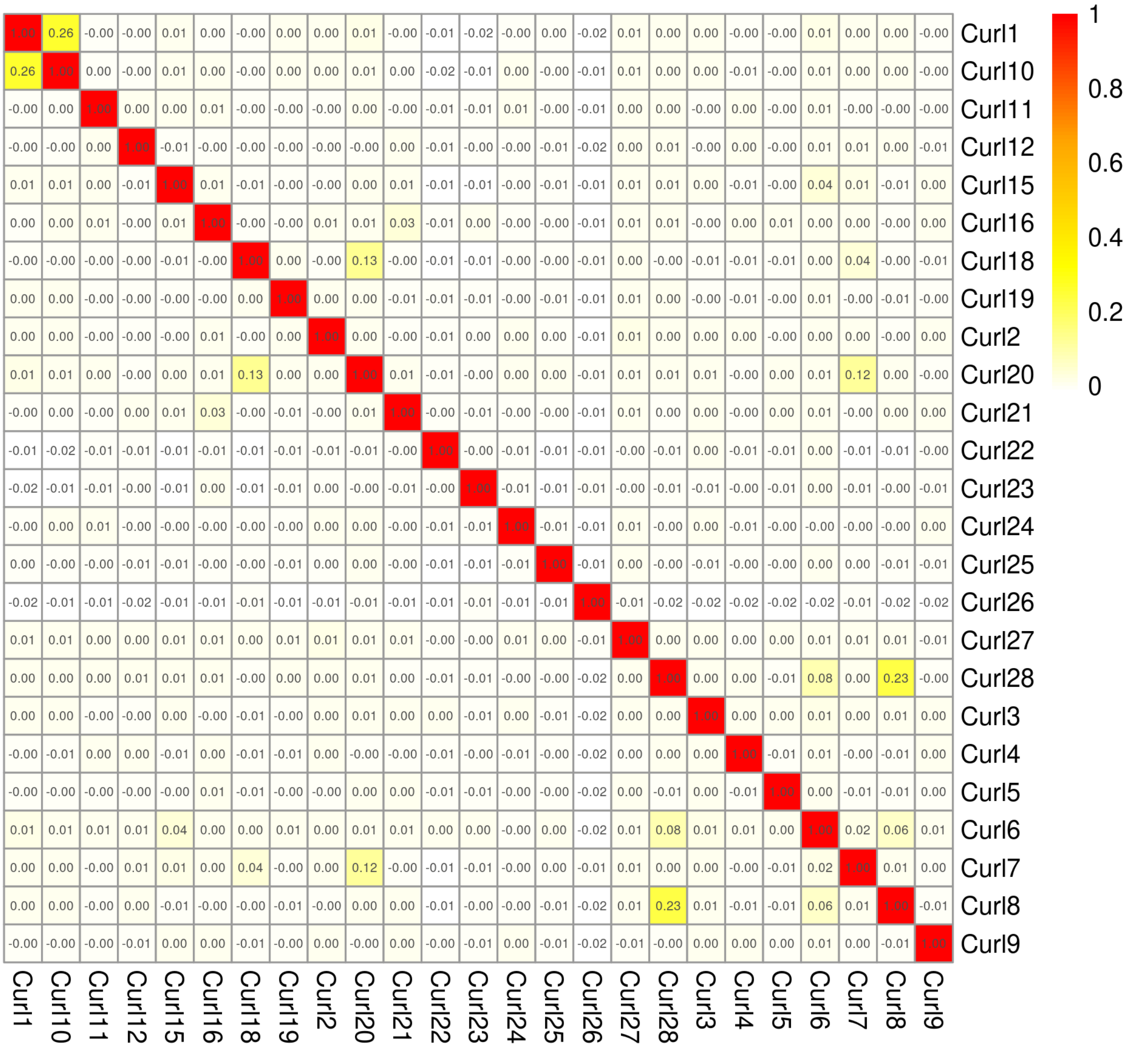

Figure S3.1: Heatmap showing pairwise kinship coefficients for 25 Irish breeding Eurasian curlews *Numenius arquata* estimated from autosomal SNPs with KING v2.2.7.

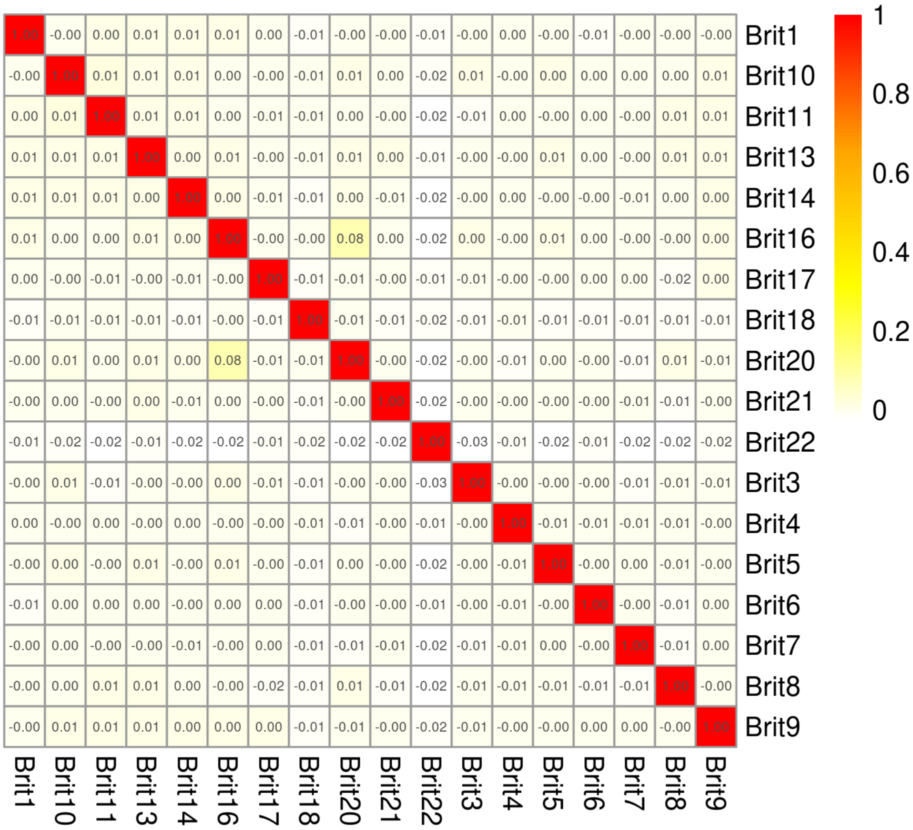

Figure S3.2: Heatmap showing pairwise kinship coefficients for 18 British breeding Eurasian curlews *Numenius arquata* estimated from autosomal SNPs with KING 2.2.7.

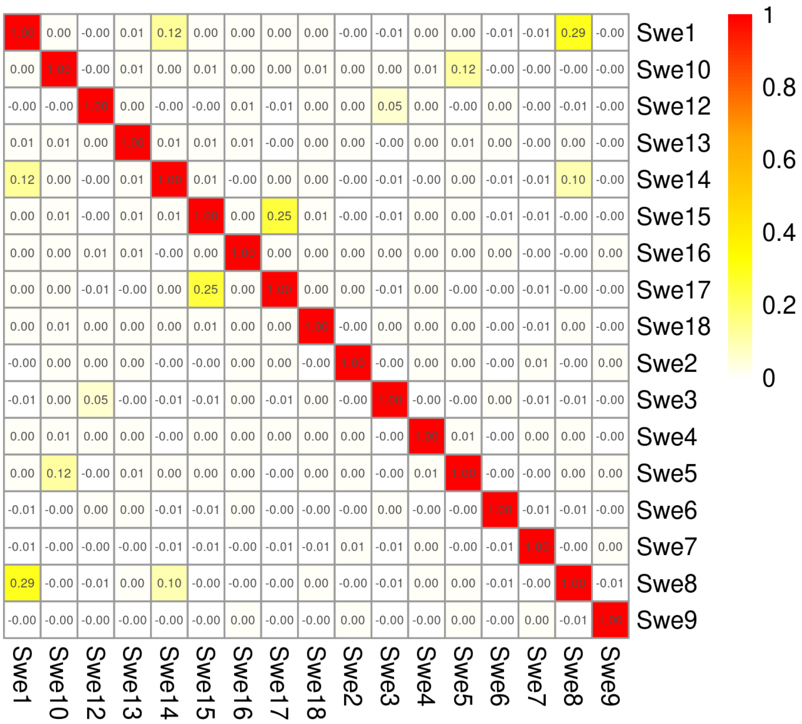

Figure S3.3: Heatmap showing pairwise kinship coefficients for 17 Swedish breeding Eurasian curlews *Numenius arquata* estimated from autosomal SNPs KING 2.2.7.

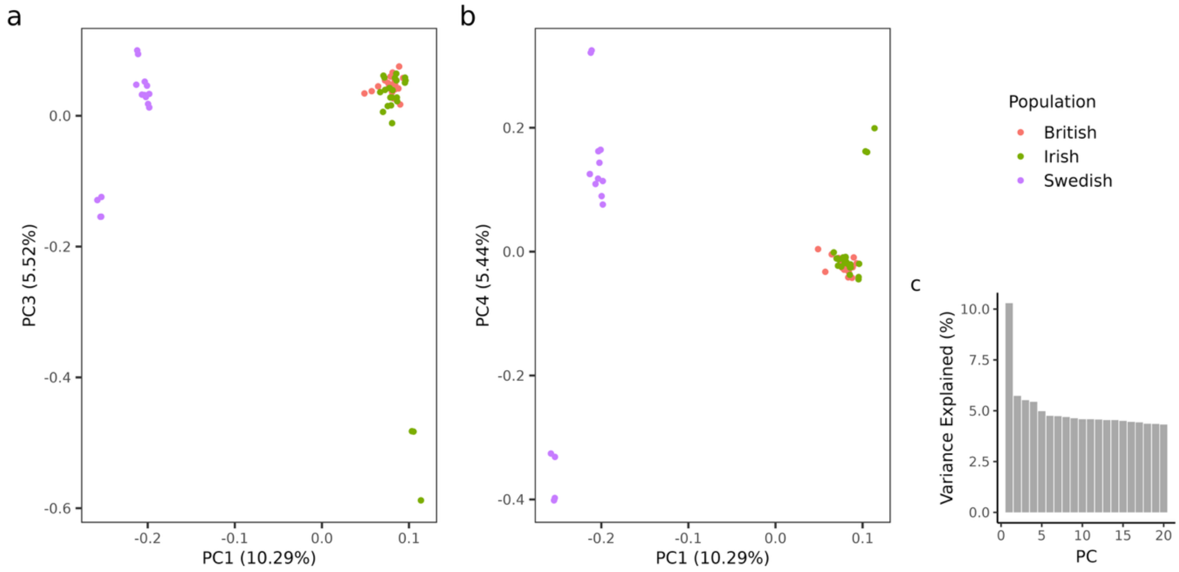

Figure S3.4: Results of the principal component analysis (PCA) for 56 Eurasian curlews (*Numenius arquata*) across three breeding populations in Ireland, Britain, and northern Sweden based on 5,523,293 SNPs. The plots show (a) PC1 versus PC3, (b) PC1 versus PC4 and (c) a histogram of the variance explained by the first 20 principal components.

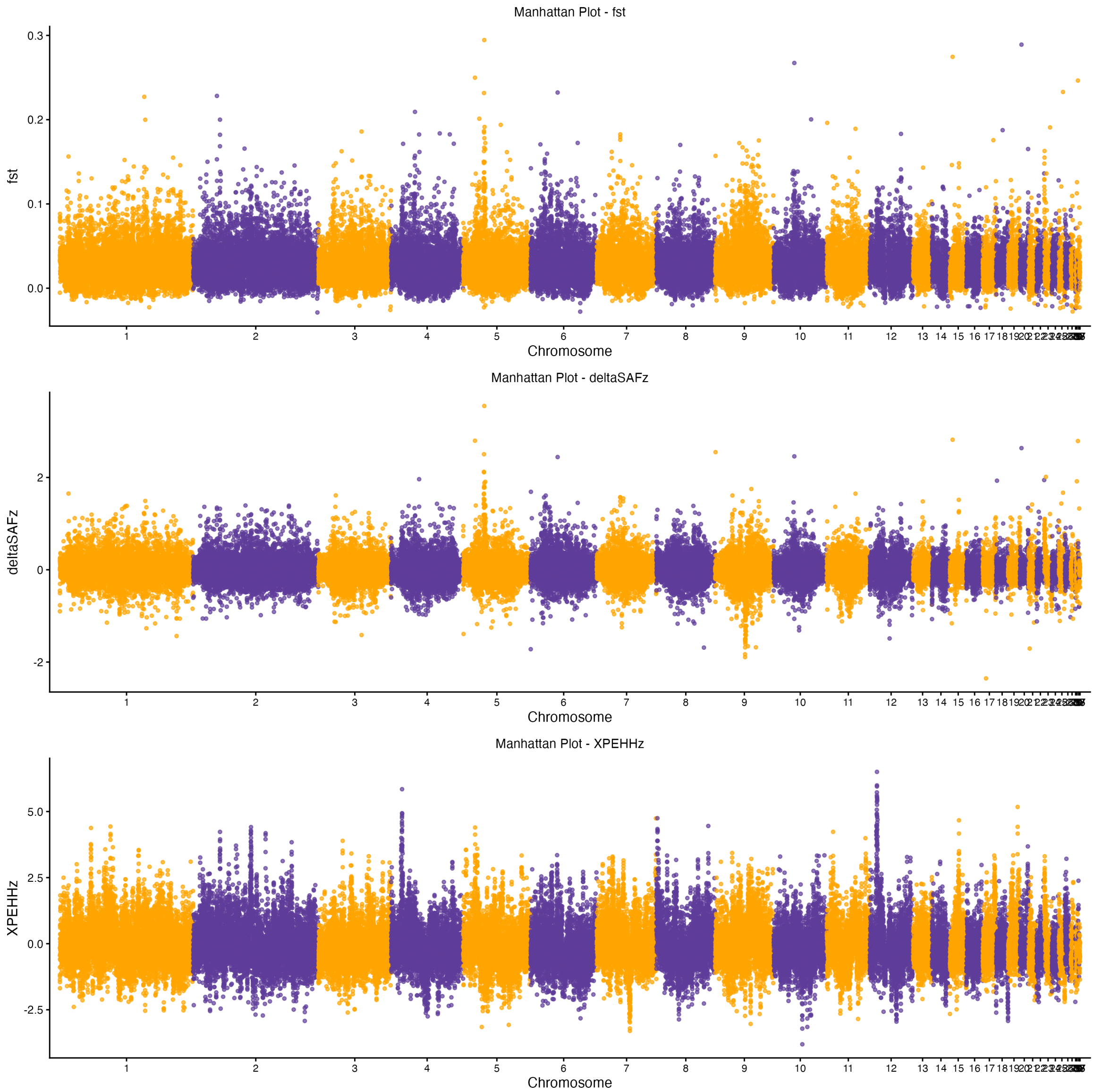

Figure S3.5: Manhattan plots of three summary statistics (F_{ST}, *ΔSAF* and *XP-EHH*) used to calculate Composite Selection Signals (CSS) for the comparison of Irish and British breeding Eurasian curlews (*Numenius arquata*) to a population in northern Sweden, using 56 whole-genome sequences.

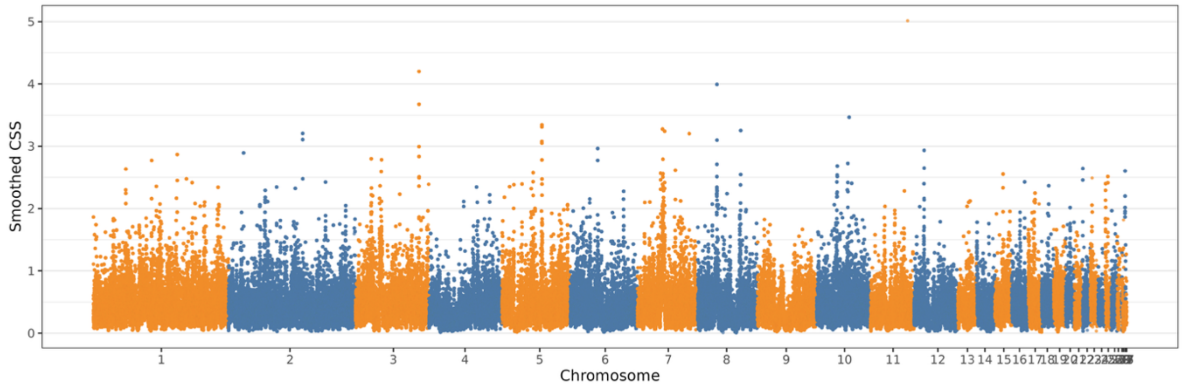

Figure S3.6: Composite Selection Signal (CSS) scores across the genome for Irish versus British Eurasian curlews (*Numenius arquata*). Peaks indicate regions under putative divergent selection based on combined F_{ST}, *XP-EHH*, and *ΔSAF* scores smoothed over 20 kb windows.

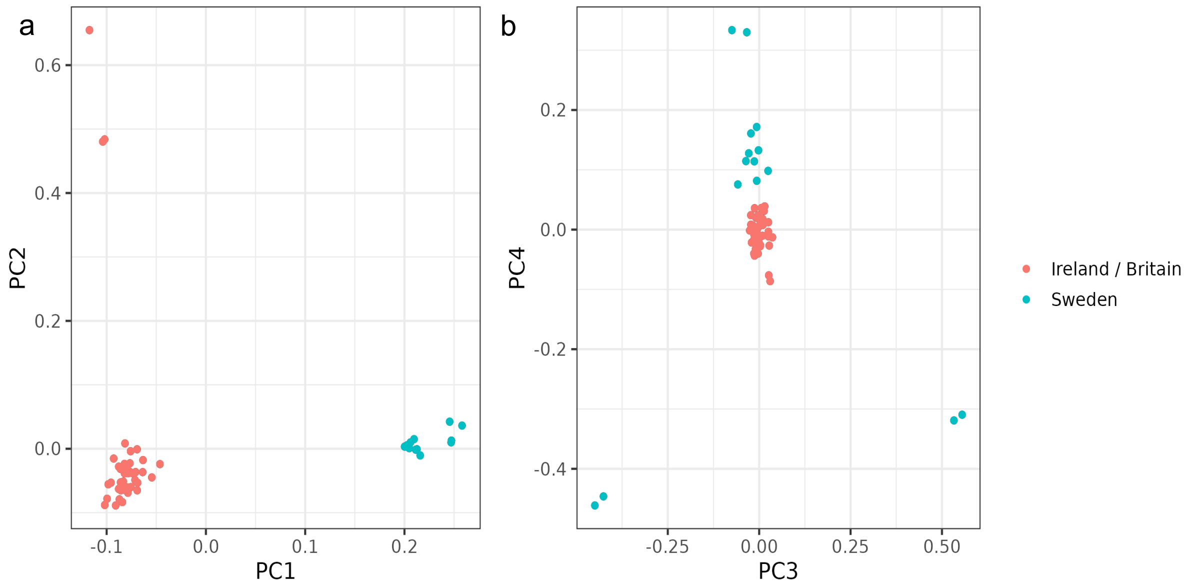

Figure S3.7: Principal component score plots for (a) PC1 and PC2, and (b) PC3 and PC4, from the pcadapt outlier method using 56 Eurasian curlews (*Numenius arquata*) across three study groups in Ireland, Britain, and northern Sweden based on 5,523,293 SNPs.

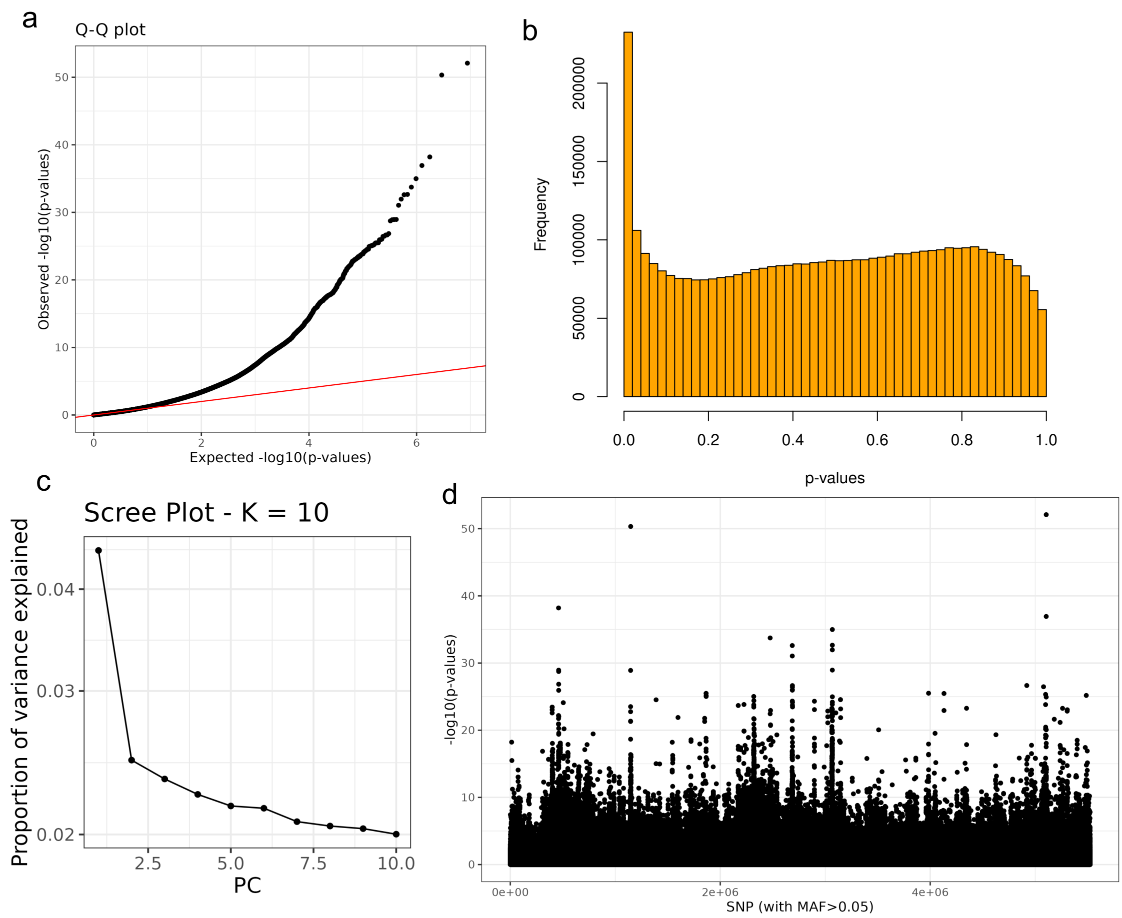

Figure S3.8: Diagnostic plots from the *pcadapt* outlier detection analysis: (a) Q-Q plot of $-\log_{10}P$ values for the SNPs, (b) histogram *P*- values frequencies, (c) scree plot showing variance explained by the first 10 principal components and (d) Manhattan plot of $-\log_{10}P$ values.

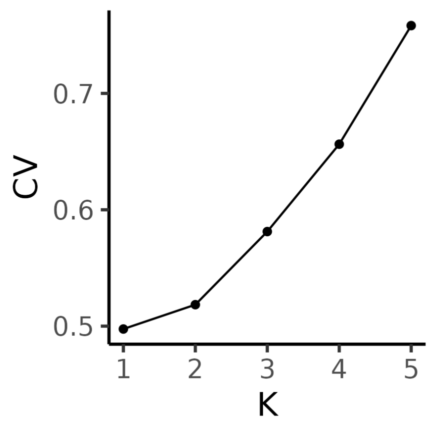

Figure S3.9: Cross-validation (CV) error from ADMIXTURE analysis for K = 1–5, where K is the assumed number of ancestral populations. CV error is lowest at K = 1 and increases with higher K.

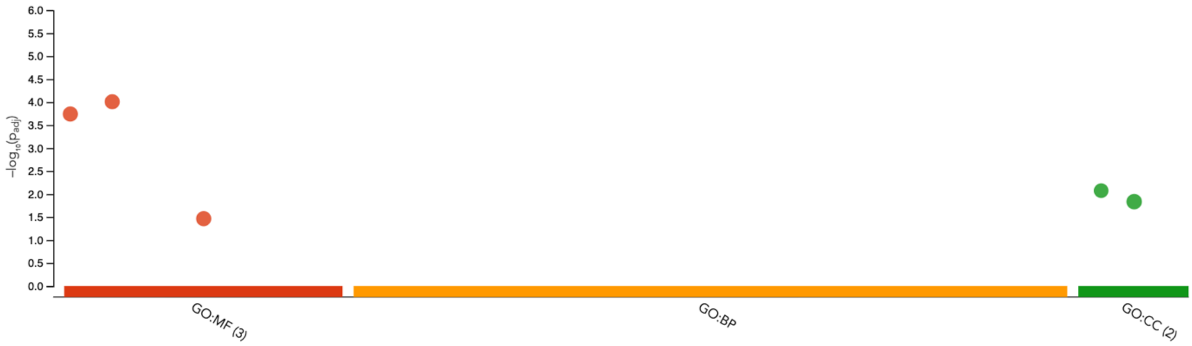

Figure S3.10: Gene Ontology (GO) overrepresentation analysis for the outlier genes (*n* = 432) identified using the R package pcadapt and Composite Selection Signal (CSS) scores for Irish/British versus Swedish Eurasian curlew (Numenius arquata) originating from populations.

Table S3.1: Summary of whole-genome resequencing batches for Eurasian curlews (*Numenius arquata*), raw read depth, and post-mapping coverage. All batches were sequenced on an Illumina NovaSeq X Plus and counts include sequences removed after quality control.

Batch no.	Library Preparation	Samples (*n*)	Date	Raw depth	Post-mapping depth
P29966	Illumina DNA PCR-Free Prep (Illumina, Inc., California, USA)	15	30/12/2023	20.4x	14.8x
P33209		18	19/11/2024	24.9x	19.4x
P34852		23	16/05/2025	30.9x	21.8x
90-1188102853	NEBNext Ultra II DNA (NEB, Ipswich, MA, USA)	10	18/04/2025	33.9x	21.3x

Table S3.2: Summary of SNP filtering steps applied to the whole-genome resequencing dataset of Eurasian curlews (*Numenius arquata*), showing the number of retained variants at each stage.

Filtering step	SNPs retained	Filtering criteria	Analysis
Initial filtering	15,953,091	QUAL < 30, MQ < 40.00, SOR > 4.000, QD < 2.00, FS > 60.000, MQRankSum < -12.5, ReadPosRankSum < -8.000	---
Autosomes	14,934,126		---
All sites autosomes	Sites: 550,160,281	QUAL < 30, MQ < 40.00, SOR > 4.000, QD < 2.00, FS > 60.000, MQRankSum < -12.5, ReadPosRankSum < -8.000	---
No MAF	7,513,513	--maxDP 40 --minDP 10 --max-missing 0.9	ROH, pixy diversity
MAC < 3	5,523,293	--mac 3 --maxDP 40 --minDP 10 --max-missing 0.9	Diversity estimates, PCA, F_{ST}
Linkage disequilibrium	782,547	$r^2 = 0.2$	ADMIXTURE
Thinning from MAC3 SNPs	816 – Ireland 818 – Britain 819 – Sweden	--thin 1000000 --max-missing 1	ONeSAMP

Table S3.3: Significant Gene Ontology (GO) overrepresentation terms of outlier genes in the Ireland/Britain vs Sweden comparison of the Eurasian curlews (*Numenius arquata*). The table lists enriched Molecular Function (MF) and Cellular Component (CC) categories, including GO term identifiers and adjusted *P*-values.

GO category	GO term	GO term ID	$P_{adj.}$ value
GO:MF	G protein-coupled peptide receptor activity	GO:0008528	0.0001
GO:MF	peptide receptor activity	GO:0001653	0.0002
GO:MF	peptide binding	GO:0042277	0.0343
GO:CC	cytosolic ribosome	GO:0022626	0.0084
GO:CC	ribosomal subunit	GO:0044391	0.0147

Table S3.4: Gene Ontology (GO) overrepresentation terms of outlier genes for the Ireland vs Britain comparison of the Eurasian curlews (*Numenius arquata*). The table lists enriched Molecular Function (MF) and Cellular Component (CC) categories, including GO term identifiers and adjusted *P*-values based on Composite Selection Signals (CSS).

GO category	GO term	GO term ID	$P_{adj.}$ value
GO:MF	volume-sensitive anion channel activity	GO:0005225	0.0500
GO:BP	positive regulation of vascular endothelial cell proliferation	GO:1905564	0.0466
GO:CC	DNA-dependent protein kinase complex	GO:0070418	0.0072
GO:CC	intracellular anatomical structure	GO:0005622	0.0170
GO:CC	cytosol	GO:0005829	0.0340
GO:CC	Ku70:Ku80 complex	GO:0043564	0.0499